# Discrepancy Problems: Definition and Context

## 1 The General Setup

A **discrepancy problem** is a class of problems in combinatorics and discrepancy theory that asks:

> *Given a finite set system, can we color the ground set with two colors so that every set in the system is as "balanced" as possible — and how balanced can we guarantee it to be?*

More precisely, one is given:

- A **ground set** $X$ (usually finite, $|X| = n$),
- A **set system** $\mathcal{F} \subseteq \mathcal{P}(X)$ — a family of subsets of $X$,
- A **two-coloring** $\chi \colon X \to \{+1, -1\}$.

For each $S \in \mathcal{F}$, the **discrepancy of** $S$ with respect to $\chi$ is the *imbalance*:

$$\operatorname{disc}_\chi(S) = \left| \sum_{x \in S} \chi(x) \right| = \big| \#\{x \in S : \chi(x) = +1\} - \#\{x \in S : \chi(x) = -1\} \big|.$$

The **discrepancy of the set system** $\mathcal{F}$ is then

$$\operatorname{disc}(\mathcal{F}) = \min_{\chi \colon X \to \{\pm 1\}} \max_{S \in \mathcal{F}} \operatorname{disc}_\chi(S).$$

That is:

- **Inner max**: over all sets in the system, how badly can a given coloring unbalance some set?
- **Outer min**: over all colorings, how well can we do in the worst case?

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## 2 What Makes It a "Problem"?

The discrepancy problem for a given set system is the task of determining (or bounding) $\operatorname{disc}(\mathcal{F})$, and often of explicitly constructing an optimal or near-optimal coloring.

There are several standard variants:

| Variant | Ground set | Set system | Coloring |
|---|---|---|---|
| **Combinatorial discrepancy** | Finite set | Arbitrary family | $\pm 1$ |
| **Spencer's problem** | $[n]$ | Arbitrary family of $m$ sets | $\pm 1$ |
| **Balancing sequences** | $[n]$ | Intervals / prefixes | $\pm 1$ (or $\{-1, 0, +1\}$) |
| **Geometric discrepancy** | Points in $\mathbb{R}^d$ | Half-spaces, boxes, etc. | $\pm 1$ |
| **Arithmetic discrepancy** | $\mathbb{Z}$ or $\mathbb{Z}_n$ | Arithmetic progressions | $\pm 1$ |

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## 3 The Key Tension

The discrepancy problem captures a fundamental tension:

- If we color too **uniformly** (e.g., half $+1$, half $-1$), small sets might still be very unbalanced (e.g., a singleton is always $\pm 1$).
- If we try to **balance every small set**, we may be forced into a coloring that unbalances some large set badly.

The discrepancy measures the *optimal worst-case tradeoff*.

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## 4 Canonical Examples

**1. The complete set system $\mathcal{P}(X)$.** As shown earlier, $\operatorname{disc}(\mathcal{P}(X)) = \lceil n/2 \rceil$. The worst sets are the monochromatic ones — any coloring has a color class of size at least $\lceil n/2 \rceil$, and that class itself is a set in the system.

**2. Intervals on a line.** For $\mathcal{F} = \{\text{intervals in } [n]\}$, the discrepancy is much smaller — it grows like $\log n$ (a classical result related to the Erdős–Szekeres theorem and to the "Steinitz lemma" in one dimension).

**3. Arbitrary families of $m$ sets (Spencer's theorem).** For any family of $m$ subsets of an $n$-element set, there exists a coloring with discrepancy $O(\sqrt{m \log(2n/m)})$. This is the celebrated **"six standard deviations suffice"** theorem of Spencer (1985).

**4. Arithmetic progressions (Erdős discrepancy problem).** For colorings of the integers and the system of all finite arithmetic progressions, the discrepancy is *unbounded* — this was the Erdős discrepancy problem, resolved affirmatively by Terence Tao in 2015.

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## 5 Summary

A discrepancy problem is:

> *The problem of finding a two-coloring of a ground set that minimizes the maximum imbalance over a prescribed family of subsets — and of determining how small that minimum can be made.*

It sits at the intersection of combinatorics, probability, geometry, and number theory, and its central question is always the same: *how evenly can we distribute two colors across all sets in a given family?*